FortiGate VLAN & Firewall Configuration Guide

Intermediate Networking — CLI Lab Reference

This guide walks through configuring VLANs and firewall rules on a FortiGate firewall using the Command Line Interface (CLI). You will create VLAN sub-interfaces, define firewall address objects for each network, and write policies to control traffic between them. Each section explains the correct order of operations, what every command does, and common mistakes to avoid.

> ■ **Order of Operations — Do This First!**
> You MUST follow this order or commands will fail:
> **1.** Create VLAN interfaces → **2.** Create firewall address objects → **3.** Create firewall policies

VLANs Used in This Lab

VLAN ID	Name	Gateway IP	Subnet	Purpose
4	Data	`10.1.1.99`	/24	Wired workstations / data traffic
6	Wifi	`10.1.12.99`	/24	Internal wireless clients
8	Servers	`192.168.2.2`	/24	Internal servers
10	Native	`10.1.0.99`	/24	Native/management trunk VLAN
11	Management	`10.1.32.99`	/24	Switch & device management (HTTP/HTTPS/Ping)
12	Optional	`10.1.20.99`	/24	Optional segment
14	Guest	`10.1.52.99`	/27	Guest Wi-Fi — Internet only, isolated

■ **Note:**
VLAN 14 (Guest) uses a /27 subnet (255.255.255.224 = 30 usable hosts) — smaller than the others intentionally to limit guest capacity.

Step 1 — Create VLAN Interfaces

What Is a VLAN Interface?

On a FortiGate, a VLAN interface is a *logical sub-interface* created on top of a physical trunk port. The trunk carries all VLAN traffic; the FortiGate tags/untags frames using the VLAN ID you specify. Each sub-interface gets its own IP address, which becomes the **default gateway** for devices in that VLAN.

How to Enter the Interface Configuration Context

```
config system interface
```

All VLAN sub-interface commands are typed inside this context. Each VLAN block starts with `edit "<name>"` and ends with `next`. Type `end` when you are done with all VLANs.

Example — VLAN 4 (Data)

```
edit "Vlan 4 Data"
set vdom "root"
set interface "Internal Trunk"
set vlanid 4
set ip 10.1.1.99 255.255.255.0
set allowaccess ping
set alias "VLAN 4 Data"
set role lan
next
```

Parameter Breakdown

`edit "Vlan 4 Data"`	Creates or edits a sub-interface named exactly this. The name must be unique.
`set vdom "root"`	Assigns this interface to the 'root' Virtual Domain. Most single-tenant setups use root.
`set interface "Internal Trunk"`	The physical parent interface that carries the VLAN trunk (802.1Q). Must already exist.

`set vlanid 4`	The 802.1Q VLAN tag number. Must match the tag configured on your switch trunk port.
`set ip 10.1.1.99 255.255.255.0`	The IP address and subnet mask for this interface — this becomes the default gateway for VLAN 4 hosts.
`set allowaccess ping`	Permits ICMP (ping) to this interface IP. Useful for testing. 'ping https http' also allows web management.
`set alias "VLAN 4 Data"`	A human-readable label shown in the GUI. Does not affect function.
`set role lan`	Marks this as a LAN-side interface. Affects available policy options in the GUI.
`next`	Saves this entry and moves to the next edit block. Do NOT forget this — without it, changes are not committed.

■ **Pro Tip:**
VLAN 11 (Management) uses `set allowaccess ping https http` — this is intentional so administrators can reach the FortiGate web UI from the management VLAN. Only add HTTP/HTTPS access to VLANs that genuinely need it.

All VLAN Interfaces — Full CLI Block

```
config system interface
# VLAN 4 — Wired Data
edit "Vlan 4 Data"
set vdom "root"
set interface "Internal Trunk"
set vlanid 4
set ip 10.1.1.99 255.255.255.0
set allowaccess ping
set alias "VLAN 4 Data"
set role lan
next
# VLAN 6 — Internal Wi-Fi
edit "Vlan 6 Wifi"
set vdom "root"
set interface "Internal Trunk"
set vlanid 6
set ip 10.1.12.99 255.255.255.0
set allowaccess ping
set alias "VLAN 6 InternWifi"
set role lan
```

```
next

# VLAN 8 — Servers
edit "Vlan 8 Servers"
set vdom "root"
set interface "Internal Trunk"
set vlanid 8
set ip 192.168.2.2 255.255.255.0
set allowaccess ping
set alias "VLAN 8 Servers"
set role lan
next

# VLAN 10 — Native
edit "Vlan 10 Native"
set vdom "root"
set interface "Internal Trunk"
set vlanid 10
set ip 10.1.0.99 255.255.255.0
set allowaccess ping
set alias "VLAN 10 Native"
set role lan
next

# VLAN 11 — Management (HTTP/HTTPS access enabled)
edit "Vlan 11 manage"
set vdom "root"
set interface "Internal Trunk"
set vlanid 11
set ip 10.1.32.99 255.255.255.0
set allowaccess ping https http
set alias "VLAN 11 Management"
set role lan
next

# VLAN 12 — Optional Segment
edit "Vlan 12 Opt"
set vdom "root"
set interface "Internal Trunk"
set vlanid 12
set ip 10.1.20.99 255.255.255.0
```

```
set allowaccess ping
set alias "VLAN 12 Opt"
set role lan
next

# VLAN 14 — Guest (/27 = only 30 hosts)
edit "Vlan 14 Guest"
set vdom "root"
set interface "Internal Trunk"
set vlanid 14
set ip 10.1.52.99 255.255.255.224
set allowaccess ping
set alias "VLAN 14 Guest"
set role lan
next

end
```

■■ Watch Out:
Common Mistake #1: Forgetting `next` between entries. Without it, the FortiGate will not save the current entry and may merge parameters from two different VLANs together.

Common Mistake #2: The VLAN ID here must match your switch's trunk configuration exactly. A mismatch means no traffic — and it can be very hard to troubleshoot.

Step 2 — Create Firewall Address Objects

What Are Address Objects?

Address objects are named representations of IP addresses, subnets, or ranges. You cannot directly type an IP address into a firewall policy on FortiGate — you must first create an address object and then reference it by name. Think of them as reusable variables for IP addresses.

How to Enter the Address Configuration Context

```
config firewall address
```

Example — Address Object for VLAN 4

```
edit "VLAN_4_Data_Net"
set comment "Network for VLAN 4 Data"
set subnet 10.1.1.0 255.255.255.0
next
```

`edit "VLAN_4_Data_Net"`	The name of the address object. Use a clear, consistent naming convention — this exact name will be referenced in policies.
`set comment "..."`	Optional but strongly recommended. Helps others (and future-you) understand what this object is for.
`set subnet 10.1.1.0 255.255.255.0`	Defines the network address and mask. Note: this is the NETWORK address (e.g., 10.1.1.0), NOT the gateway IP (10.1.1.99).

■■ Watch Out:
Common Mistake #3: The system auto-generates UUIDs (`set uuid ...`). Do NOT try to set the UUID manually — the FortiGate will reject it with an error. Simply omit the uuid line when entering new address objects.

■ Note:
The Guest VLAN (14) uses subnet `10.1.52.96 255.255.255.224` — notice the network address is .96, not .0. This is because a /27 block starting at .96 covers .96–.127. Always calculate your network address carefully for non-/24 subnets.

All VLAN Address Objects — Full CLI Block

```
config firewall address
edit "VLAN_4_Data_Net"
set comment "Network for VLAN 4 Data"
set subnet 10.1.1.0 255.255.255.0
```

```
next
edit "VLAN_6_Wifi_Net"
set comment "Network for VLAN 6 Internal WiFi"
set subnet 10.1.12.0 255.255.255.0
next
edit "VLAN_8_Servers_Net"
set comment "Network for VLAN 8 Servers"
set subnet 192.168.2.0 255.255.255.0
next
edit "VLAN_10_Native_Net"
set comment "Network for VLAN 10 Native"
set subnet 10.1.0.0 255.255.255.0
next
edit "VLAN_11_Management_Net"
set comment "Network for VLAN 11 Management"
set subnet 10.1.32.0 255.255.255.0
next
edit "VLAN_12_Opt_Net"
set comment "Network for VLAN 12 Optional"
set subnet 10.1.20.0 255.255.255.0
next
edit "Vlan 14 Guest"
set comment "Guest Network — /27 limited"
set subnet 10.1.52.96 255.255.255.224
next
end
```

Step 3 — Create Firewall Policies

What Is a Firewall Policy?

A firewall policy defines what traffic is allowed to flow between two interfaces. Without a policy, all inter-VLAN traffic is **denied by default**. Each policy specifies a source interface, destination interface, source address, destination address, and an action (accept or deny). Policies are matched top-down — the first matching policy wins.

How to Enter the Policy Configuration Context

```
config firewall policy
```

Policy Type 1 — Inter-VLAN (Stateful, No NAT)

Use this pattern when you want traffic to flow between two internal VLANs, such as allowing the Data VLAN to reach the Wi-Fi VLAN. NAT is **disabled** because both sides are private networks — there is no need to translate addresses.

```
edit 4
set name "Vlan4toVlan6-Stateful"
set srcintf "Vlan 4 Data"
set dstintf "Vlan 6 Wifi"
set srcaddr "VLAN_4_Data_Net"
set dstaddr "VLAN_6_Wifi_Net"
set action accept
set service "ALL"
set nat disable
set schedule "always"
next
```

Policy Type 2 — VLAN to Internet (NAT Enabled)

Use this pattern for any VLAN that needs Internet access. The destination interface is `wan1` and **NAT must be enabled** — the FortiGate translates the private VLAN addresses to its public WAN IP before sending traffic upstream.

```
edit 5
set name "Guest-Internet"
set srcintf "Vlan 14 Guest"
set dstintf "wan1"
set srcaddr "all"
set dstaddr "all"
set action accept
set service "ALL"
```

```
    set nat enable
    set schedule "always"
    next
```

Policy Parameter Breakdown

`edit`	The policy ID number. Must be unique. Policies are evaluated in ascending ID order — lower IDs are checked first.
`set name "..."`	A descriptive name. No functional effect, but essential for readability and troubleshooting.
`set srcintf "..."`	The source interface — where traffic is coming FROM. Must match an interface name created in Step 1.
`set dstintf "..."`	The destination interface — where traffic is going TO. Use 'wan1' for Internet-bound traffic.
`set srcaddr "..."`	The source address object. Must match a name created in Step 2, or use the built-in 'all'.
`set dstaddr "..."`	The destination address object. Usually 'all' for Internet policies; specific objects for inter-VLAN.
`set action accept`	Allows matching traffic. Use 'deny' to explicitly block (useful for logging dropped traffic).
`set service "ALL"`	Applies to all TCP/UDP/ICMP services. Can be narrowed to specific ports (e.g., 'HTTP', 'HTTPS', 'DNS').
`set nat enable`	Enables Source NAT (SNAT). Required for Internet access. The FortiGate substitutes its WAN IP as the source.
`set nat disable`	Disables NAT. Use for inter-VLAN policies where both sides are private RFC1918 addresses.
`set schedule "always"`	Applies the policy 24/7. You can create time-based schedules to restrict access to certain hours.

■■ Watch Out:
Common Mistake #4: Referencing an interface or address object that does not exist yet. The CLI will accept the command but the policy will be broken — always create interfaces (Step 1) and address objects (Step 2) before writing policies (Step 3).

Common Mistake #5: Duplicate policy names in the source document (e.g., several policies all named 'Guest-Net2'). Names do not have to be unique on FortiGate, but using duplicates makes it very difficult to identify policies during troubleshooting. Always use descriptive, unique names.

■ **Pro Tip:**
After creating policies, use `show firewall policy` to verify them, or `diagnose firewall iprope show 100004 <policy_id>` to inspect a specific policy's match counters. You can also use the GUI Policy & Objects view to drag and reorder policies visually.

Quick Reference & Common Mistakes Summary

CLI Cheat Sheet

`config system interface`	Enter VLAN interface configuration mode
`config firewall address`	Enter address object configuration mode
`config firewall policy`	Enter firewall policy configuration mode
`edit "" / edit`	Create or open an existing entry for editing
`next`	Save current entry and move to next (do not forget this!)
`end`	Exit the current config context and save all changes
`show firewall policy`	Display all configured firewall policies
`show firewall address`	Display all configured address objects
`show system interface`	Display all configured interfaces including VLANs
`get system interface physical`	Show physical interface status and link state

Common Mistakes At a Glance

#	Mistake	Why It Matters
#1	**Forgetting 'next'**	Each edit block must end with 'next'. Without it, your config is not committed and parameters may bleed into the next entry.
#2	**VLAN ID mismatch**	The vlanid on the FortiGate must match the VLAN tag on the switch trunk port. Even a single digit off means no traffic.
#3	**Manually setting UUID**	UUIDs are assigned by the system. If you paste existing configs that include 'set uuid ...', remove those lines first.
#4	**Wrong order of operations**	Policies reference interfaces and address objects. If either doesn't exist yet, the policy will fail or be misconfigured silently.
#5	**Duplicate policy names**	FortiGate allows duplicate names — but this makes management a nightmare. Always use unique, descriptive policy names.
#6	**NAT on inter-VLAN policies**	Inter-VLAN traffic between private subnets does NOT need NAT. Enabling NAT here causes routing issues and unexpected source IPs.

#7	**Network vs host address in address objects**	Address objects use the network address (e.g., 10.1.1.0/24), not the gateway IP (10.1.1.99). Using the gateway IP will cause the policy to match only that single host.

This guide covers the fundamental CLI workflow. For advanced topics — VDOM segmentation, SD-WAN policies, SSL inspection, or IPS profiles — refer to the Fortinet documentation portal at docs.fortinet.com.